

UNIT I

CONDUCTING MATERIALS

ALE

SYLLABUS:

Classical free electron theory – merits and limitations - expression for electrical conductivity –Fermi-Dirac distribution function – Density of energy states – Energy bands in solids –Electron effective mass (Concept only) - concept of holes – Computed Tomography scan.

Introduction:

Low resistive materials are known as conducting materials. They have high electric and thermal conductivities. The conducting property of a solid is due to the number of valence electrons, which become free electron when they are detached from the parent atom.

Classification of conducting materials:

Based on their conductivity conducting materials are classified into three types.

- Zero resistivity materials
- Low resistivity materials
- High resistivity materials

Electron theory of solids

The Electron theory of solid explains the structure and properties of solids through their electronic structure. It is applicable to all solids both metals and nonmetals.

The three main theories developed for metals are

- (i) Classical free electron theory, which is a macroscopic theory obeying classical laws.
- (ii) Quantum free electron theory, which is a microscopic theory obeying quantum laws.
- (iii) Zone theory (or) Band theory is also a microscopic theory which is based on the energy bands of solids.

CLASSICAL FREE ELECTRON THEORY

- This theory was developed by Drude and Lorentz in 1900.
- According to this theory, a metal consists of electrons which are free to move about in the crystal.
- It is a macroscopic theory and obeys the laws of classical mechanics.
- The total energy of the electron is equal to its kinetic energy.
- This theory obeys Maxwell-Boltzman statistics which permits all free electrons to gain energy

QUANTUM FREE ELECTRON THEORY

- It is a microscopic theory proposed by Somerfield in 1928.
- This theory obeys quantum mechanical concepts and Fermi-Dirac statistics to the free electrons in the metals.
- Quantum free electron theory permits only a few electrons to gain energy.

- It explains that the electrons move in a constant potential.

ZONE THEORY (OR) BAND THEORY

- Band theory developed by Bloch in 1928.
- It explains the electrons move in a periodic potential field.
- This theory explained the Lattice concept of holes, origin of Band gap and effective mass of electrons. These are the special features of this theory of solids.
- This theory also explains the mechanism of super conductivity based on band theory.

CLASSICAL FREE ELECTRON THEORY

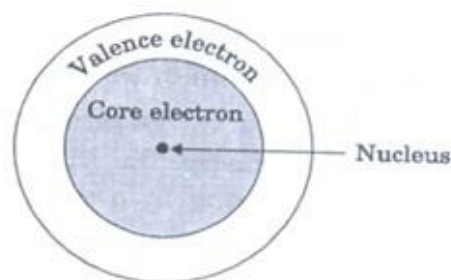
This theory was proposed by Drude and Lorentz in the year 1900. According to this theory free electrons present in the metal is responsible for electrical conduction. This theory obeys all the laws of classical mechanics. Here the free electrons are assumed to move in a constant potential.

An atom consists of a central nucleus with positive charge. The negatively charged electrons are surrounding the nucleus. The electrons in the closed shell are called as core electrons and those in the outermost shell are called as valence electrons core electrons are strongly attracted by the nucleus but the valence electrons are free electrons. Drude assumed that the electrons in a metal are free to move and they virtually form an electron gas. The free electrons move randomly in all possible directions just like gas molecule. Lorenz predicted that kinetic theory of gases could be applicable to the free electron gas.

When the atoms are brought closer to form a metal, the valence electrons get detached and move freely through the metal. Hence, they are called free or conducting electrons. The concentration of free electrons is 10^{28} m^{-3} . These free electrons are charged are assigned with a mean free path, a mean collision time and an average speed.

POSTULATES OF CLASSICAL FREE ELECTRON THEORY

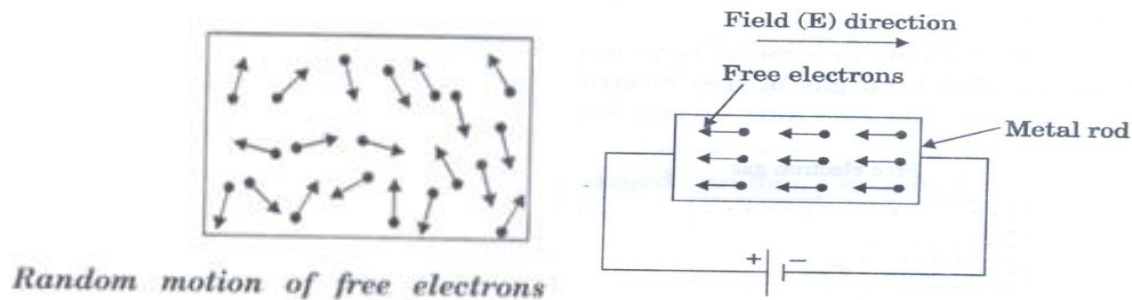
1. A metal is made up of atoms with nucleus at its centre, and the electrons are revolving around the nucleus in different orbits or shells. The electrons present in the shells which are closer to the nucleus are called as core electrons and electrons in the outermost shell are called as valence electrons.



Atom (Drude model)

2. When the valence electrons are detached from the orbit they move freely throughout the volume of the metal. Now these electrons are called free electrons.

3. In the absence of an electric field, the free electrons move freely in random manner and they collide with each other. This type of collision is called as elastic collision.
4. When an external field is applied, the electrons acquire some amount of energy and move towards the positive potential with some constant velocity called as drift velocity(v_d).



Movement of free electrons with applied electric field

5. Since the electrons are assumed to be perfect gas (free electrons behave like gas molecules in a container), they obey the laws of **kinetic theory of gases**. Therefore, the free electrons are assigned with mean free path, collision time and average velocity.

MERITS OF CLASSICAL FREE ELECTRON THEORY

1. It is used to verify ohm's law.
2. It is used to explain the electrical and thermal conductivity of metals.
3. It is used to derive Wiedmann-Franz law.
4. It is used to explain the optical properties of metals.

DRAWBACKS OF CLASSICAL FREE ELECTRON THEORY

1. Classical theory states that all free electrons absorb the supplied energy. But quantum theory states that only a few electrons absorb the supplied energy.
2. The electrical conductivity of semiconductors and insulators cannot be explained by this theory.
3. The photo electric effect, Compton effect and blackbody radiation cannot be explained by this theory.
4. According to this theory, the ratio $\frac{K}{\sigma T}$ is a constant at all temperatures. But it is found that it is not constant at low temperatures.
5. By this theory, the value of specific heat capacity of a metal is $4.5R$. But the experimental value is $3R$, where R is a universal gas constant.
6. The susceptibility of a paramagnetic material is inversely proportional to temperature. But experimental result shows that Paramagnetism of a metal is independent of temperature.
7. Moreover, Ferromagnetism cannot be explained by this theory.

Basic Definitions: -

Drift Velocity(v_d):

The average velocity acquired by free electrons in a particular direction by the application of an electric field is called drift velocity (v_d). *i.e.*, $v_d = \frac{\lambda}{\tau_c}$

Mean free path(λ):

The average distance travelled by a free electron between any two successive collisions is known as mean free path(λ). It is the product of drift velocity (v_d) and collision time(τ_c). *i.e.*, $\lambda = v_d \times \tau_c$

Collision time(τ_c):

The average time taken by a free electron between any two successive collisions is known as collision time. *i.e.*, $\tau_c = \frac{\lambda}{v_d}$

Relaxation time(τ):

The average time taken by a free electron to reach its equilibrium position from its disturbed position due to the application of an external electric field is called relaxation time. It is approximately equal to 10^{-14} second. (Almost $\tau = \tau_c$)

Wiedemann Franz law:

The ratio of thermal conductivity to electrical conductivity of a metal is directly proportional to the absolute temperature of the metal. This ratio is constant for all metals at a given temperature.

$$\frac{K}{\sigma} \propto T$$

$$\frac{K}{\sigma} = LT$$

Where L is a proportionality constant known as Lorentz number.

The value of L is $2.44 \times 10^{-8} \text{ W}\Omega\text{K}^{-2}$

Thermal conductivity:

Thermal conductivity (K) of a material is defined as the amount of heat conducted (Q) per unit time (t) through the material having unit area of cross section (A) per unit temperature gradient $\left(\frac{dT}{dx}\right)$.

$$K = \frac{Q}{t \times A \times \left(\frac{dT}{dx}\right)}$$

If area of cross section $A = 1 \text{ m}^2$ and $t = 1 \text{ second}$, then

$$K = \frac{Q}{\left(\frac{dT}{dx}\right)} \quad \text{and} \quad Q = K \frac{dT}{dx} \quad \text{or} \quad K \frac{dT}{\lambda} \quad (\text{if the distance} = \lambda)$$

ELECTRICAL CONDUCTIVITY OF A METAL (DERIVATION):

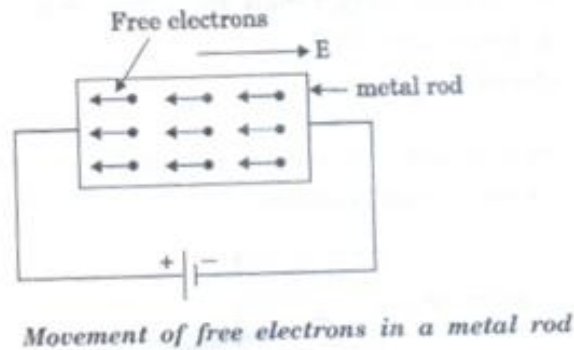
The amount of electrical charges (Q) conducted per unit time (t) across unit area (A) of a conductor per unit applied electric field (E). It is denoted by σ and it is given by,

$$\sigma = \frac{Q}{tAE} = \frac{J}{E} \quad ; \quad \text{where } J = \frac{Q}{tA}$$

Where, σ – electrical conductivity, t – time, A – Area of cross section,

E – Applied electric field, J – Current density;

J can be written in different forms; $J = \sigma E$, $J = nev_d$



In the absence of an electric field, the free electrons move freely throughout the volume of the metal in random manner. When an external field (E) is applied, the electrons acquire some amount of energy and moves towards the positive potential. As a result, the electron acquires a constant velocity called as drift velocity(v_d).

If ' E ' is the applied electric field, then the force acting on each electron,

$$F = eE \quad \text{where, } e \rightarrow \text{charge of electron.} \quad \text{--- (1)}$$

But according to Newton's second law of motion,

$$F = \text{Mass of the electron (m)} \times \text{acceleration (a)}$$

$$F = ma \quad \text{--- (2)}$$

From eqn (1) and (2) we get, $ma = eE$

$$a = \frac{eE}{m} \quad \text{--- (3)}$$

We know average drift velocity of electron is v_d and collision time is τ ($\tau = \tau_c$)

$$\text{General formula for acceleration is,} \quad a = \frac{\text{velocity}}{\text{time}};$$

$$\text{here acceleration,} \quad a = \frac{v_d}{\tau} \quad \text{--- (4)}$$

$$\text{From (4)} \quad v_d = a\tau \quad \text{--- (5)}$$

$$\text{Substituting (3) in (5) We get,} \quad v_d = \frac{eE}{m}\tau \quad \text{--- (6)}$$

Current density in terms of drift velocity can be written as

$$J = nev_d \quad \text{--- (7)}$$

Substitute v_d from (6) in (7) We get,

$$J = n e \left(\frac{e E \tau}{m} \right) \quad \text{--- (8)}$$

$$J = \frac{n e^2 \tau}{m} E \quad \text{--- (9)}$$

$$\frac{J}{E} = \frac{n e^2 \tau}{m} \quad \text{--- (10)}$$

From Ohm's law, Current density J is expressed as

$$J = \sigma E \quad \text{and} \quad \sigma = \frac{J}{E} \quad \text{--- (11)}$$

Substituting the (11) in (10) we get,

$$\sigma = \frac{n e^2 \tau}{m} \quad \text{--- (12)}$$

The equation (12) represents electrical conductivity of the metal.

FERMI DIRAC STATISTICS:

Fermi – Dirac statistics deals with particles having half integral spin. Particles like electrons obeying the FD statistics are called fermions. The expression which gives the distributions of electrons among the various energy levels as a function of temperature is known as Fermi distribution function.

The probability $F(E)$ of an electron occupying a given energy level at absolute zero temperature is called Fermi – Dirac distribution function.

It is given by,

$$F(E) = \frac{1}{1 + e^{\frac{(E-E_F)}{kT}}}$$

Where, $E \rightarrow$ Energy of the level whose occupancy is considered;

$E_f \rightarrow$ Fermi energy (energy of Fermi level)

$k \rightarrow$ Boltzmann's constant ;

$T \rightarrow$ Absolute temperature

The probability value of $F(E)$ lies between 0 and 1.

If $F(E) = 1 \rightarrow$ the considered energy level is occupied by electrons

If $F(E) = 0 \rightarrow$ the considered energy level is not occupied by electrons (Energy level is vacant)

If $F(E) = 0.5$ or $\frac{1}{2}$

$\rightarrow$ there is a 50% chance for the electron occupying in that considered energy level)

Fermi energy:

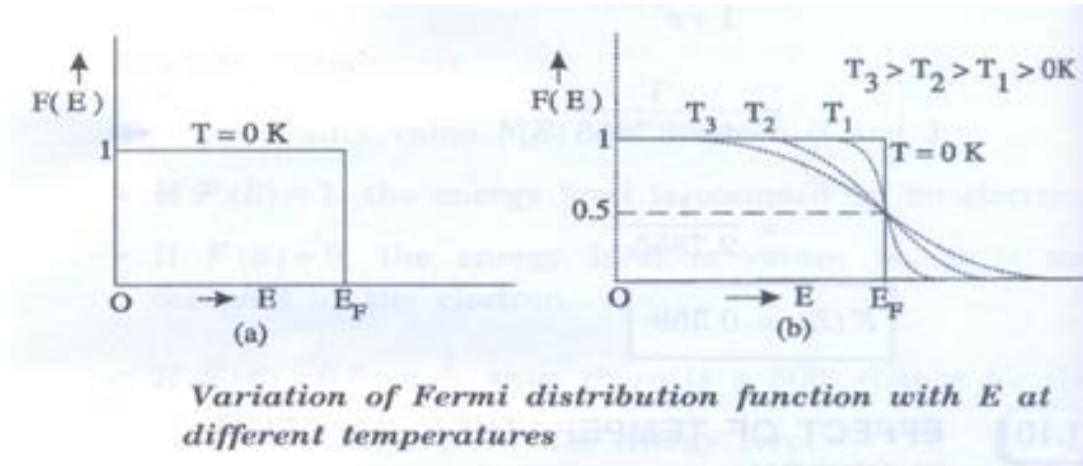
It is the energy of the state at which the probability of electron occupation is $\frac{1}{2}$ at any temperature above 0K. It is the maximum energy of filled states at 0K.

Fermi energy level:

It is the energy level at which the probability of electron occupation is $\frac{1}{2}$ or 50 % at any temperature above 0 K. It is also the energy level of maximum energy of the filled states at 0 K

EFFECT OF TEMPERATURE ON FERMİ FUNCTION

The variation of F.D distribution function with temperature is shown in fig.



Case 1: At $T=0K$ for $E < E_F$ (Considered level is below the Fermi level)

$$F(E) = \frac{1}{1 + e^{\frac{-ive}{0}}} = \frac{1}{1 + e^{-\infty}} = \frac{1}{1 + 0} = 1 \quad \because e^{\infty} = \infty \text{ and } e^{-\infty} = 0$$
$$\therefore F(E) = 1$$

There is 100% chance for the electrons to occupy the energy level below Fermi energy level.

Case 2: At $0K$ for $E > E_F$ (Considered level is above the Fermi level)

$$F(E) = \frac{1}{1 + e^{\frac{+ive}{0}}} = \frac{1}{1 + e^{\infty}} = \frac{1}{1 + \infty} = \frac{1}{\infty} = 0 \quad ; \quad \text{Hence, } F(E) = 0$$

There is 0% chance for the electrons to occupy the energy level above Fermi energy level.

Case 3: At any temperature other than $0K$ and $E = E_F$

With increase of temperature $T > 0K$ Fermi function $F(E)$ varies with E as Shown in Fig. (b)

$$F(E) = \frac{1}{1 + e^{\frac{0}{kT}}} = \frac{1}{1 + e^0} = \frac{1}{(1 + 1)} = \frac{1}{2} = 0.5 \quad \text{Because, } E = E_f. \text{ So, } E - E_f = 0$$
$$\therefore F(E) = 0.5$$

There is 50% chance for the electrons to occupy Fermi energy level.

Case 4:

At very high temperature when $kT \gg E_F$ or $T \rightarrow \infty$ the electrons lose their quantum mechanical character and the Fermi distribution function reduces to classical Boltzmann distribution function.

USES OF FERMİ DISTRIBUTION FUNCTION

1. It gives the probability of occupation for a given energy level at a given temperature.
2. It is very useful in the calculation of number of free electrons per unit volume at given temperature.
3. It is very used to find the Fermi energy of a metal.

DENSITY OF ENERGY STATES

DEFINITION

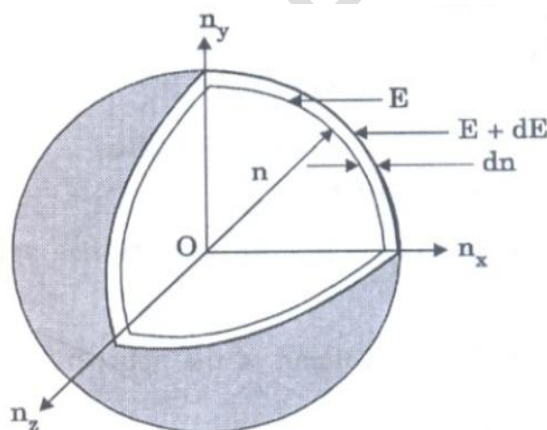
Density of states is defined as number of available electron states (energy states) per unit volume in an energy interval E and $E+dE$. It is denoted by $Z(E)dE$. It is given by,

$$Z(E)dE = \frac{\text{No. of energy states in an energy interval } E \text{ and } E + dE}{\text{Volume of the metal piece}} \quad \text{--- (1)}$$

$$Z(E)dE = \frac{N(E)dE}{V} \quad \text{--- (2)}$$

TO FIND THE DENSITY OF ENERGY STATES (DERIVATION)

Let us consider a sphere of metal with three quantum numbers n_x, n_y, n_z as three co-ordinate axes. The radius vector ' n ' is drawn from the origin ' O ' to a point on the sphere, its value can be calculated using the formula $n^2 = n_x^2 + n_y^2 + n_z^2$. All the points on this sphere having same energy ' E '. This sphere is divided into many shells. Each shell is considered as one energy state. Let us also consider another one sphere, which is little bit bigger than older one. Radius of this sphere is ' $n+dn$ '. All the points on this sphere having same energy ' $E+dE$ '. This sphere is also divided into many shells. Each shell is considered as one energy state. We have to find out number of energy levels available in between these two spheres $N(E)dE$.



$$\text{No. of energy states within sphere of radius 'n'} = \frac{4}{3}\pi n^3 \quad \text{--- (3)}$$

$$\text{No. of energy states within sphere of radius 'n + dn'} = \frac{4}{3}\pi (n + dn)^3 \quad \text{--- (4)}$$

The quantum numbers (n_x, n_y and n_z) can have only positive integer values. It will have positive values at $\frac{1^{th}}{8}$ part (octant) of the sphere. So we have to consider only $\frac{1^{th}}{8}$ part of the spherical volume.

$$\begin{aligned} \text{No. of energy states within one octant } \left(\frac{1^{th}}{8}\right) \text{ of sphere with radius 'n'} &= \frac{1}{8} \times \frac{4}{3}\pi n^3 \\ &= \frac{1}{6}\pi n^3 \quad \text{--- (5)} \end{aligned}$$

$$\text{No. of energy states within one octant } \left(\frac{1^{th}}{8}\right) \text{ of sphere with radius 'n + dn'} = \frac{1}{8} \times \frac{4}{3}\pi (n + dn)^3$$

$$= \frac{1}{6} \pi (n + dn)^3 \quad \text{--- (6)}$$

No. of energy states in an energy interval E and $E + dE$ (Subtracting (5) from (6))

$$N(E) dE = \frac{1}{6} \pi (n + dn)^3 - \frac{1}{6} \pi n^3$$

$$N(E) dE = \frac{1}{6} \pi [(n + dn)^3 - n^3]$$

$$\because (a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$N(E)dE = \frac{1}{6} \pi (n^3 + 3n^2dn + 3ndn^2 + dn^3 - n^3) \quad \text{--- (7)}$$

Since 'dn' is very small the higher orders of 'dn' (dn^2 terms & dn^3 terms) can be neglected.

$$N(E) dE = \frac{\pi}{6} \times 3n^2 dn$$

$$N(E)dE = \frac{\pi}{2} n \times n dn \quad \text{--- (8)}$$

The energy of an electron in a cubical metal piece of sides 'a' is given by,

$$E = \frac{n^2 h^2}{8ma^2} \quad \text{--- (9)}$$

$$n^2 h^2 = 8ma^2 E \quad (\text{by cross multiplication})$$

$$n^2 = \frac{8ma^2 E}{h^2} \quad \text{--- (10)}$$

$$n = \left[\frac{8ma^2 E}{h^2} \right]^{\frac{1}{2}} \quad \text{--- (11)}$$

$$\text{Differentiating equation (10) with respect to 'n' we get, } 2n dn = \frac{8ma^2 dE}{h^2} \quad \text{--- (12)}$$

$$n dn = \frac{8ma^2 dE}{2h^2} \quad \text{--- (13)}$$

Substituting in (11) and (13) in (8) we get, $N(E)dE$

$$= \frac{\pi}{2} \left(\frac{8ma^2 E}{h^2} \right)^{\frac{1}{2}} \times \left(\frac{8ma^2 dE}{2h^2} \right) \quad \text{--- (14)}$$

$$N(E)dE = \frac{\pi}{4} \left(\frac{8ma^2}{h^2} \right)^{\frac{1}{2}} \times \left(\frac{8ma^2}{h^2} \right)^1 E^{\frac{1}{2}} dE$$

$$N(E)dE = \frac{\pi}{4} \left(\frac{8ma^2}{h^2} \right)^{\frac{3}{2}} E^{\frac{1}{2}} dE \quad \text{--- (15) } (\because a^m \times a^n = a^{m+n})$$

Pauli's exclusion principle states that "Two electrons of opposite spin can occupy each state". Hence, the number of energy states available for electron occupancy is given by, (Eqn. (15) should be multiplied by '2')

$$N(E)dE = 2 \times \frac{\pi}{4} \left(\frac{8ma^2}{h^2} \right)^{\frac{3}{2}} E^{\frac{1}{2}} dE \quad \text{--- (16)}$$

$$N(E)dE = \frac{\pi}{2} \left(\frac{8ma^2}{h^2} \right)^{\frac{3}{2}} E^{\frac{1}{2}} dE \quad \text{--- (17)}$$

$$N(E)dE = \frac{\pi}{2h^3} 8^{\frac{3}{2}} m^{\frac{3}{2}} a^3 E^{\frac{1}{2}} dE \quad \text{--- (18)} \left(\because (h^2)^{\frac{3}{2}} = h^3; (a^2)^{\frac{3}{2}} = a^3 \right)$$

$$N(E)dE = \frac{\pi}{2h^3} 8 \times 8^{\frac{1}{2}} \times m^{\frac{3}{2}} \times a^3 \times E^{\frac{1}{2}} dE \quad \text{--- (19)} \left(\because 8^{\frac{3}{2}} = 8^1 \times 8^{\frac{1}{2}} \right)$$

$$N(E)dE = \frac{4\pi}{h^3} 2^{\frac{3}{2}} m^{\frac{3}{2}} a^3 E^{\frac{1}{2}} dE \quad \left(\because 8^{\frac{1}{2}} = \sqrt{8} = \sqrt{2 \times 2 \times 2} = 2\sqrt{2} = 2^1 \times 2^{\frac{1}{2}} = 2^{\frac{3}{2}} \right)$$

$$N(E)dE = \frac{4\pi}{h^3} (2m)^{\frac{3}{2}} a^3 E^{\frac{1}{2}} dE \quad \text{--- (20)}$$

We know, Density of energy states $Z(E)dE = \frac{N(E)dE}{V} \quad \text{--- (2)}$

Volume of the metal cube, $V = a^3 \quad \text{--- (21)}$

Substitute (20) and (21) in (2) we get, $Z(E)dE = \frac{\frac{4\pi}{h^3} (2m)^{\frac{3}{2}} a^3 E^{\frac{1}{2}} dE}{a^3}$

$$Z(E)dE = \frac{4\pi}{h^3} (2m)^{\frac{3}{2}} E^{\frac{1}{2}} dE \quad \text{--- (22)}$$

This is the expression for density of charge carriers in the energy interval E and E+dE.

ENERGY BANDS IN SOLIDS:

Energy band theory of solids plays a very important role in determining whether a solid is a conductor, insulator or a semiconductor.

In an isolated atom all the electrons are very closely packed with the central positive nucleus and revolve around various orbits. The number of electron at the outermost orbit is called valence electrons. In the outermost orbits, the attractive force between the nucleus and electrons will be very loss, So that the electrons can be easily detected from the nucleus. These detected electros from the outer most orbits are called free electrons. But as far as the innermost orbits are concerned, the electrons are tightly bounded with positive nucleus and hence they are termed as bound electrons.

A convenient way of representing the energy of different orbits is called energy levels. The larger the orbit of an electron, the greater is its energy and higher is the energy level.

The collection of such a large number of energy levels is called energy band. In an energy band, allowed energy levels are almost continuous. These energy bands are separated by ranges of energies that have no allowed energy levels. These regions are known as forbidden band or energy gaps.

The band formed by a series of energy levels containing valence electrons is known as valence band. The next higher permitted band in a solid in the conduction band. The electrons occupying this band are known as conduction electrons. The conduction and valence bands are separated by a gap is known as forbidden energy gap.

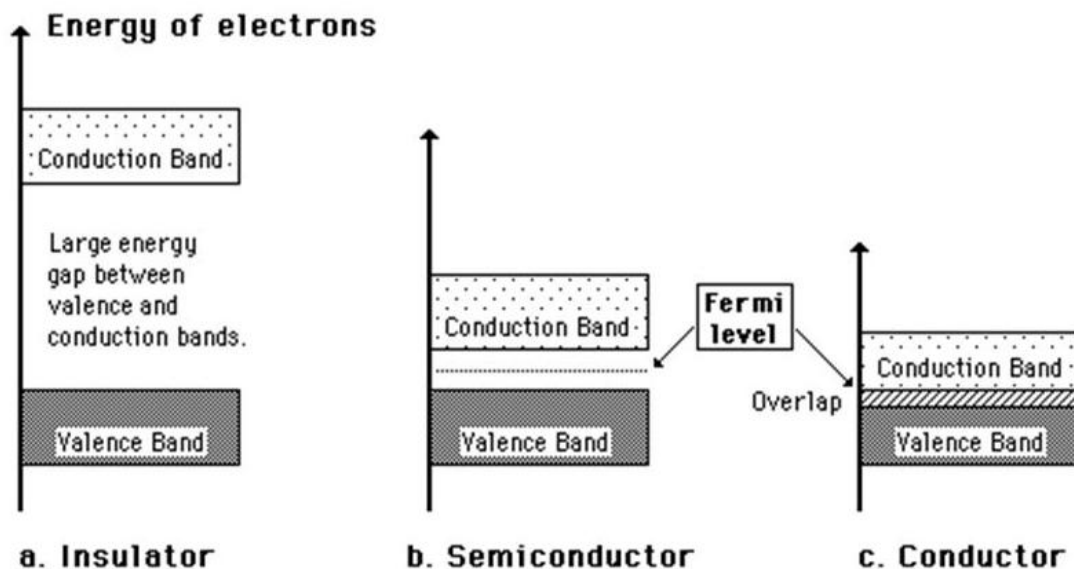
On the bases of energy bands, solids can be classified is to insulators, semiconductors and conductors.

Insulators

- In case of insulators, the E_g is very large.
- Due to this, electron cannot jump from valance band to conduction band.
- In insulators, the valence electrons are band very tightly to their parent atoms.
- E_g : Diamond $-E_g=6V$ & Glass $E_g=10eV$
- No free electrons are available to conduct electricity.
- When a very large electric energy is supplied an elector maw jump across he forbidden energy gap. this explain why certain insulators became conductors at high temperature

Semiconductors:

- In semiconductors, the E_g is very small.
- Ge ($E_g=0.7eV$) and Si ($E_g=1.1eV$) are the best example of semiconductors.
- Semiconductors is a material where electrical conductor lies between insulators & conductors
- At $0k$ there are no free electrons in the conduction band & valence band is completely filled.
- When a small amount of energy is applied, the electrons can easily jump from valence band to conduction band, since E_g is very small.



Conductors:

- In conductors, there is no E_g .
- The valence band & conduction band overlap each other.
- Metals have a high on account of the presence of large no of free electrons.
- The total I in the conductor is simply the flow of electrons
- E_g cu, siag.

EFFECTIVE MASS OF ELECTRON:

In general, however, the value of effective mass depends on the purpose for which it is used, and can vary depending on a number of factors. For electrons or electron holes in a solid, the effective mass is usually stated in units of the rest mass of an electron, m_e (9.11×10^{-31} kg).

The mass acquired by an electron when it is accelerated in a periodic potential is called effective mass of an electron. It is denoted by m^* . When an electron is accelerated by an electrical or magnetic field in a periodic potential, the mass of the electron is not a constant. But it varies with respect to the field applied. This varying mass is called effective mass (m^*).

CONCEPT OF HOLE

The effective mass plays an important role in conduction process of semiconductors and insulators since they have full or almost filled valence bands. We can find that the effective mass m^* is negative near the zone edges of almost filled valence bands. Physically speaking the electrons in these regions are accelerated in a direction opposite to the direction of the applied field. This is called the negative mass behaviour of the electrons. The electrons with the negative effective mass is considered as a new entity having the same positive mass of that of an electron but with positive charge. The new entity is given the name “hole”. The holes are not real particles like electrons or positrons, but it is a way of looking at the negative mass behaviour of the electrons near the zone edge. We look upon the motion of the effective negative mass electrons as the motion of positive holes or positive vacant sites in a nearly full band and allow the electrons in the band to carry the current. The positive hole conduction and effective negative electron mass conduction are in equilibrium. The calculations made on the hole picture appear to be more convenient and hence the hole concept is advantageously retained.

- The electrons with the negative effective mass is considered as a new entity having the same positive mass of that of an electron but with positive charge and is given as name “hole”.
 - The holes are not real particles like electrons or positrons, but it is a way of looking at the negative mass behavior of the electrons near the zone edge.
 - Several phenomena like Hall effect, Thomson effect, etc are on the basis of hole concept.
-

COMPUTED TOMOGRAPHY (CT) SCAN:

Introduction:

Computed Tomography (CT) scan is an advanced medical imaging technique used to obtain detailed cross-sectional images of the internal parts of the human body. Unlike conventional X-ray imaging, which produces a two-dimensional projection, CT combines X-rays, detectors and computer processing to produce detailed images of tissues, organs, bones and blood vessels. A CT scanner acquires X-ray measurements from different angles around the patient. A computer then reconstructs these measurements to generate cross-sectional slices (tomographic images). Multiple slices can also be combined to form a three-dimensional image.

Block Diagram:



Construction:



The main components of a CT scanner are:

(i) X-ray Source

- A high-energy X-ray tube produces a narrow beam of X-rays.
- The X-ray source rotates around the patient during scanning.

(ii) Detector Array

- X-ray detectors are positioned opposite to the X-ray source.
- They detect the intensity of X-rays after passing through the patient's body.
- Modern CT scanners use a large number of detectors to acquire images rapidly.

(iii) Gantry

- The gantry is a circular structure containing the X-ray tube and detector system.
- These components rotate around the patient.

(iv) Patient Table

- The patient lies on a movable motorized table.
- The table moves gradually through the gantry during scanning.

(v) Data Acquisition System

- It collects electrical signals from the detectors.
- The signals are converted into digital data for processing.

(vi) Computer System

- A high-speed computer processes the measured X-ray data.
- It reconstructs cross-sectional and three-dimensional images.

(vii) Display System

- The reconstructed images are displayed on a monitor.
- The images can be stored electronically for diagnosis and further analysis.

Working:

The working principle of a CT scan is based on the differential absorption of X-rays by different tissues of the human body. During the scanning process, the patient is placed on a movable table inside the CT scanner, and an X-ray tube rotates around the patient, emitting X-rays from different directions. As the X-rays pass through the body, different tissues absorb different amounts of radiation depending on their density and composition. The transmitted X-rays are detected by a detector array, which converts the received X-ray intensity into electrical signals. These signals are then converted into digital data by the data acquisition system and processed by a computer. Using measurements obtained from multiple angles, mathematical reconstruction techniques are employed to generate detailed cross-sectional images of the body. These images are displayed as two-dimensional slices, and multiple slices can be combined to produce a three-dimensional representation of the internal structures. The image contrast depends on the X-ray attenuation coefficient of the tissues; bones absorb X-rays strongly and therefore appear bright, soft tissues exhibit intermediate contrast, while air absorbs very little X-ray radiation and appears dark.

Advantages:

1. Produces detailed cross-sectional images of internal organs.
2. Provides better visualization of soft tissues than conventional X-ray imaging.
3. Can produce three-dimensional images.
4. Useful for imaging the brain, chest, abdomen, bones, and blood vessels.
5. CT scanning is generally fast, making it useful in emergency situations.
6. Helps detect internal injuries and abnormalities accurately.
7. Provides precise information about the location, size and shape of abnormalities.
8. Can assist doctors in planning certain diagnostic and treatment procedures.
9. Digital images can be stored and transmitted easily.
10. Provides images without the overlap of structures commonly seen in ordinary X-ray radiographs.

Disadvantages:

1. CT uses ionizing X-ray radiation.
2. Radiation exposure is generally higher than that from a single conventional X-ray examination.
3. CT scanners are expensive to install and maintain.
4. The equipment requires sophisticated computer systems and trained personnel.
5. Some examinations may require a contrast agent, which may not be suitable for every patient.
6. Image quality can be affected by patient movement.
7. CT may not always be the preferred technique when another imaging method can provide the required information without ionizing radiation.
8. Repeated imaging should be avoided unless medically necessary.

Applications:

CT scanning has numerous applications in medicine and engineering-related biomedical fields.

(i) Brain Imaging

CT is used to examine the brain and identify abnormalities such as:

- Head injuries
- Brain abnormalities
- Internal bleeding
- Structural changes

(ii) Bone Imaging

CT provides detailed images of:

- Complex bone structures
- Fractures
- Joints
- Spine

(iii) Chest Imaging

CT is used to examine:

- Lungs
- Chest structures
- Thoracic abnormalities

(iv) Abdominal Imaging

CT helps visualize internal organs such as:

- Liver
- Kidneys
- Pancreas
- Intestines

(v) Blood Vessel Imaging

Specialized CT techniques can be used to obtain images of blood vessels and study vascular structures.

(vi) Cancer Diagnosis and Treatment Planning

CT images can help determine the location and extent of tumors and can also support treatment planning and monitoring.

(vii) Emergency Medicine

CT is particularly useful for rapidly examining patients with suspected internal injuries.

(viii) Image-Guided Procedures

CT images can assist medical professionals in accurately locating internal structures during certain diagnostic and treatment procedures.

.....

PROBLEMS:

1. Calculate the drift velocity of the free electrons (with a mobility of $3.5 \times 10^{-3} \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$) in copper for an electric field strength of 0.5 V/m.

Solution:

Given data: Mobility of free electrons (μ) = $3.5 \times 10^{-3} \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$

Electric field strength of copper $E = 0.5 \text{ V/m}$

$$\text{Drift Velocity } V_d = \mu \times E$$

$$= 3.5 \times 10^{-3} \times 0.5$$

$$V_d = 0.00175 \text{ m/s}$$

2. The resistivity of copper is $1.72 \times 10^{-8} \Omega\text{m}$. Calculate the mobility of electrons in copper. Given $n = 10.41 \times 10^{28} \text{ m}^{-3}$.

Solution:

Given data: $\rho = 1.72 \times 10^{-8} \Omega\text{m}$

$\sigma = 5.81 \times 10^6 \text{ mho/m}$

We know that $\sigma = ne\mu$

$$\mu = \frac{\sigma}{ne} = \frac{5.81 \times 10^6}{10.41 \times 10^{28} \times 1.602 \times 10^{-19}}$$

$$\mu = 3.48 \times 10^{-3} \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$$

3. The thermal and electrical conductivities of copper at 20°C are $390 \text{ Wm}^{-1} \text{ K}^{-1}$ and $5.87 \times 10^7 \Omega^{-1} \text{ m}^{-1}$ respectively. Calculate Lorentz number.

Solution:

Given data: Electrical conductivity of copper $\sigma = 5.87 \times 10^7 \Omega^{-1} \text{ m}^{-1}$

Thermal conductivity of copper $K = 390 \text{ Wm}^{-1} \text{ K}^{-1}$

Temperature $T = 20^\circ \text{C} = (20 + 273) = 293 \text{ K}$

$$\text{We know that Lorentz number, } L = \frac{K}{\sigma T} = \frac{390}{5.87 \times 10^7 \times 293}$$

$$L = 2.267 \times 10^{-8} \text{ W}\Omega \text{ K}^{-2}$$

4. A silver wire has a resistivity of $1.34 \times 10^{-8} \Omega\text{m}$ at room temperature for an electric field of 50 volt/m. The relaxation time of electron is $3.97 \times 10^{-14} \text{ s}$. Calculate drift velocity of electrons.

Solution:

Given data: $\rho = 1.34 \times 10^{-8} \Omega\text{m}$

$E = 50 \text{ volt/m}$

$\tau = 3.97 \times 10^{-14} \text{ s}$

$$V_d = \frac{eE\tau}{m} \text{ m/s} = \frac{1.6 \times 10^{-19} \times 50 \times 3.97 \times 10^{-14}}{9.1 \times 10^{-31}}$$

$$V_d = 0.349 \text{ m/s}$$

5. Calculate the Fermi function for an energy $2K_B T$ above the Fermi energy.

Solution: Given data: $(E - E_F) = 2K_B T$

$$F(E) = \frac{1}{1 + e^{\frac{(E - E_F)}{k_B T}}} \\ = 0.119$$